

Shengjun Kris Dong

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Research Interests

Ph.D. candidate in AI for systems and hardware–software co-design for AI infra; Specialized in architecture-aware code optimization, performance engineering for large scale model deployment, agentic flow for system optimization.

Education

University of California, Berkeley

2022 – Present

Ph.D. in Electrical Engineering and Computer Sciences

University of California, Berkeley

2021 – 2022

Master in Electrical Engineering and Computer Sciences

University of Massachusetts Amherst

2016 – 2020

Bachelor in Computer Science, Mathematics and Finance

Skillset

Programming:

C/C++, CUDA, Java, Scala, Python, Warp, JavaScript, Bash, MATLAB

Machine Learning:

MLIR, IREE, PyTorch, TensorFlow, ONNX, R, SciPy, NumPy,

GPU Programming:

Triton / CUTLASS / CUDA Kernel Optimization, CUDA, OpenCL, GPU Profiling (i.e. Nsys/NCU)

Toolchains and Compilers:

LLVM/Clang, RISC-V GNU Toolchain, GCC (RVV backends), CMake, MLIR, Zephyr RTOS, ROS

Selected Journals and Conferences

Under Submission

2026

Shengjun Kris Dong, et al, “XPU-RT: Feedback-Driven Realtime Deployment of Robotic Workloads on Heterogeneous SoCs.” *Under Submission*

ArXiv

[\[Link\]](#)

Shengjun Kris Dong, ..., Siva Kumar Sastry Hari, Christos Kozyrakis, et al, “KernelBlaster: Continual Cross-Task CUDA Optimization via Memory-Augmented In-Context Reinforcement Learning.”

IISWC'25

11/2025

IEEE International Symposium on Workload Characterization

[\[Link\]](#)

Shengjun Kris Dong, et al, “Characterizing and Optimizing Real-Time Optimal Control for Embedded SoCs.” (**IISWC Distinguished Artifact Award**)

ISCA'23

06/2023

IEEE International Symposium on Computer Architecture

[\[Link\]](#)

Dima Nikiforov, **Shengjun Kris Dong**, et al. “RoSÉ: A Hardware-Software Co-Simulation Infrastructure Enabling Pre-Silicon Full-Stack Robotics SoC Evaluation.” (**ISCA Distinguished Artifact Award**)

RA-L'25

01/2025

IEEE Robotics and Automation Letters

[\[Link\]](#)

Jingqi Li, Donggun Lee, Jaewon Lee, **Kris Shengjun Dong**, Somayeh Sojoudi, Claire J. Tomlin. “Certifiable Deep Learning for Reachability using a new Lipschitz Continuous Value Function.”

LAD'24

06/2024

IEEE International Workshop on LLM-Aided Design

[\[Link\]](#)

Charles Hong, Sahil Bhatia, Altan Haan, **Shengjun Kris Dong**, Dima Nikiforov, Alvin Cheung, Sophia Shao. “LLM-Aided Compilation for Tensor Accelerators.”

JSSC'26

02/2026

Journal of Solid-State Circuits

[\[Link\]](#)

Vikram Jain, Daniel Grubb, Jerry Zhao, Kevin Anderson, Ken Ho, **Shengjun Kris Dong**, et al. “Cygnus: A Heterogeneous Octa-Core RISC-V Vector Processor for Digital Signal Processing Applications.”

Work Experience

PhD Research Engineer, Decision Intelligence

10/2026 – Present

NVIDIA

- Developing and implementing agentic LLM-based methods for generating and optimizing GPU code for NVIDIA cuOpt, a GPU-accelerated optimization library for large-scale decision intelligence problems spanning linear, quadratic, mixed-integer, and vehicle-routing optimization solvers and APIs.
- Building automated workflows that use LLM agents to iteratively generate, evaluate, and refine implementations of performance-critical GPU kernels, improving execution efficiency and scalability for problems with large numbers of variables and constraints.

Machine Learning Engineer, Apple Intelligence

05/2026 – Present

Apple

- Developed a hardware-aware agentic AI framework for optimizing real-time multimodal VLM/VLA-style models targeting deployment across iPhone, Mac, and wearable devices.
- Analyze multi-modal AI workloads (VLM and LLM) on cloud infrastructure through systems-level performance engineering, spanning serving efficiency, latency optimization, and workload characterization.
- Built an in-context reinforcement learning workflow that uses runtime feedback to identify, prioritize, and prune system-level optimization knobs, enabling autonomous discovery of architecture-specific strategies for memory, scheduling, and compute optimization.

PhD Research Intern, Architecture Research Group

05/2025 – 12/2025

NVIDIA

- Developed a novel agentic AI framework for CUDA kernel optimization techniques that autonomously adapts to runtime feedback, achieving architecture-specific performance gains.
- Analyze and optimize compiler-lowered GPU kernels, PTX and hardware-level instruction behavior in a closed-loop profiling environment, to identify instruction-level and memory-system optimization opportunities.

Lecturer

05/2023 – 09/2023

ROAR Academy, UC Berkeley

- Taught lecture series “Introduction to Embedded Systems and Edge ML” for ROAR Academy at UC Berkeley.
- Designed the curriculum, organized lab sessions on embedded systems, machine learning, and robotic prototyping and coordinated a full-day student competition in robotics and VR/AR.

System Architect

12/2021 – 12/2022

Tesla, Inc.

Designed and evaluated closed-loop control systems for long-distance teleoperation, analyzing trade-offs between latency, energy consumption, and system reliability under real-world operating conditions.

Project Manager

09/2020 – 09/2021

Amazon, Inc.

- Led the design and deployment of *Alexa for Warehouse*, a task-oriented LLM agent for automating complex operational workflows, supporting multi-turn reasoning, and tool invocation in a warehouse operation setting.
- Evaluated the agent-based automation of engineering and analysis pipelines, improving order distribution efficiency by 1.19× in production environments.

Research Projects

XPU-RT: Resource Orchestration and Scheduling for Multi-Model Real-Time Systems

[\[Link\]](#)

PI: Sophia Yakun Shao

- Developed a lightweight compilation flow by integrating and optimizing ML compiler stacks for multi-model AI on heterogeneous multi-accelerator systems to minimize latency under dependency and resource constraints..
- Implemented profile-guided scheduling and code generation to reduce contention and latency, closely mirroring GPU kernel and shader optimization workflows.
- Built a lightweight compilation flow that lowers high-level representations in PyTorch into low-level C/assembly and optimizes compiler-generated code paths, enabling direct inspection, transformation, and optimization.

In-context Reinforcement Learning for Agentic CUDA Code Optimization

NVIDIA Research

Pls: Christos Kozyrakis, Siva Hari, Sana Damani

[\[Link\]](#)

- Designed and developed CUDAPilot, a hardware-aware agentic AI framework for CUDA code optimization.
- Developed a Prompt-as-Policy framework using memory-augmented in-context reinforcement learning, enabling prompts to evolve adaptively during inference to discover and compose novel optimization strategies.
- Achieved a 2.21x speedup over PyTorch performance and a faster convergence rate across diverse GPUs.

LLM-aided Code Compilation for Tensor Accelerators

[\[Link\]](#)

Pls: Sophia Yakun Shao, Alvin Cheung

Leveraged LLMs to compile high-level code to tensor accelerators and developed a two-phase “Plan-then-Implement” workflow with beam search, enabling efficient code generation across systolic arrays, vector cores, and GPUs while adapting to new ISAs through prompt engineering.

Performance Optimization via Hardware-software Co-design

[\[Link\]](#)

Pls: Sophia Yakun Shao, Christopher W. Fletcher

- Conducted a comprehensive design space exploration to enable embodied AI and robotic applications on a customized systolic array and vector cores, via architectural modifications, batching, and kernel reformulation.
- Discovered novel code optimization techniques for real-time optimal control workloads, and validated a hardware-in-the-loop evaluation on a fabricated RISC-V multi-core vector SoC, achieving a 3.71x speedup on control algorithms and a 27% system-level power reduction in end-to-end robotic applications.

Pre-silicon SoC Simulation Infrastructure

[\[Link\]](#)

Pls: Yakun Sophia Shao, Borivoje Nikolic

- Built a simulation infrastructure enabling closed-loop, pre-silicon evaluation of robotics-focused SoCs.
- Co-developed a full-stack environment combining robotic dynamics, sensing, and environment simulation with an FPGA-accelerated cycle-exact RTL simulation of the SoC.
- Implemented a robust packet-level synchronization scheme for simulated sensor and actuator data flows across heterogeneous simulators.

Workshop Papers and Tapeout Papers

IISWC'26

2026

IISWC Tutorial

[\[Link\]](#)

Dima Nikiforov, **Shengjun Kris Dong**, et al, “Building, Tracing, and Optimizing End-to-End ML and Robotic Workloads on Heterogeneous RISC-V SoCs.” *IISWC Tutorial 2026, Boulder, Colorado*.

GHC'26

10/2026

GHC Poster Session

Shengjun Kris Dong, et al. “BAST: Adaptive Scheduling of Real-Time Robotic Workloads on Heterogeneous SoCs.” *Academic Poster Session, Grace Hopper Celebration 2026, Anaheim, California*.

ISCA'26

08/2026

ISCA'26 OSCAR Workshop Paper

[\[Link\]](#)

Agustin N. Coppari Hollmann, **Shengjun Kris Dong**, et al, “Merlin: Agile Compiler Infrastructure for HW/SW Co-Design.” *OSCAR Workshop at ISCA 2026*

VLSI'26

06/2026

Symposium on VLSI Technology & Circuits

[\[Link\]](#)

Vikram Jain, Hasan Nazim Genc, Coleman Hooper, Minh Nguyen, Rahul Kumar, Rohan Kumar, Di Wang, Kevin Anderson, Yufeng Chi, **Shengjun Kris Dong**, et al. “*Sirius: A Dual-Chiplet System for Multimodal Embodied AI with Heterogeneous RVV Cores, Dense and Sparse Accelerators.*”

MICRO'22

11/2022

Dima Nikiforov, **Shengjun Kris Dong**, et al. “DryDock: A Co-Simulation Platform Enabling System-Level Pre-Silicon Evaluation of Robotics SoCs.” *55th Annual IEEE/ACM International Symposium on Microarchitecture, 1st RoboARCH Workshop, IEEE, 2022*.

ILLIXR Consortium

04/2022

Dima Nikiforov, **Shengjun Kris Dong**, et al. “A Co-Simulation Platform Enabling System-Level Pre-Silicon Evaluation of Robotics SoCs.” *ILLIXR Consortium, Oral Presentation, University of Illinois at Urbana-Champaign, 2022*.